

Topic 1: Basic Linux Commands

Sl No.	Comm and	Purpose	Example
1	pwd	Display the current working directory	pwd
2	clear	Clear the terminal screen	clear
3	history	Display previously executed commands	history
4	whoami	Display the current logged-in username	whoami
5	date	Display the current system date and time	date
6	cal	Display the calendar	cal
7	hostname	Display the system hostname	hostname
8	uname -a	Display operating system information	uname -a
9	exit	Exit the terminal	exit
		<pre> mokshitha_putti@MokshithaPutti:~\$ pwd /home/mokshitha_putti mokshitha_putti@MokshithaPutti:~\$ history 1 cal 2 whoami 3 ls -l 4 mkdir ANU 5 ls 6 cd ANU 7 history 8 nano test1.txt 9 ls 10 ls -l 11 ls -a 12 cd 13 du 14 pwd 15 man 16 history 17 mkdir 18 find 19 locate 20 touch 21 cat 22 ls 23 mkdir s2 24 ls 25 touch example.txt 26 cd s2 27 touch example.txt 28 echo "Hello world" >> example.txt 29 echo "Hello world" >> example.txt 30 cat example.txt </pre>	

		<pre> 106 history 107 nano Week1.c 108 gcc Week1.c -o Week1 109 ./Week1 110 nano week12.c 111 gcc week12.c -o week12 112 ./week12 113 nano week 13.c 114 nano week13.c 115 gcc week13.c -o week13 116 ./week13 117 nano week14.c 118 gcc week14.c -o week14 119 ./week14 120 nano week15.c 121 gcc week15.c -o week15 122 ./week15 123 nano week 16.c 124 nano week16.c 125 gcc week16.c -o week16 126 ./week16 127 nano week17.c 128 gcc week17.c -o week17 129 ./week17 130 nano week18.c 131 gcc week18.c -o week18 132 nano week18.c 133 gcc week18.c -o week18 134 nano week19.c 135 gcc week19.c -o week19 136 ./week19 137 pwd 138 history mokshitha_putti@MokshithaPutti:~\$ whoami mokshitha_putti mokshitha_putti@MokshithaPutti:~\$ date Tue Aug 4 14:15:04 UTC 2026 mokshitha_putti@MokshithaPutti:~\$ cal August 2026 Su Mo Tu We Th Fr Sa 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31 mokshitha_putti@MokshithaPutti:~\$ hostname MokshithaPutti mokshitha_putti@MokshithaPutti:~\$ uname -a Linux MokshithaPutti 6.6.87.2-microsoft-standard-WSL2 Thu Jun 5 18:30:46 UTC 2025 x86_64 GNU/Linux mokshitha_putti@MokshithaPutti:~\$ </pre>	
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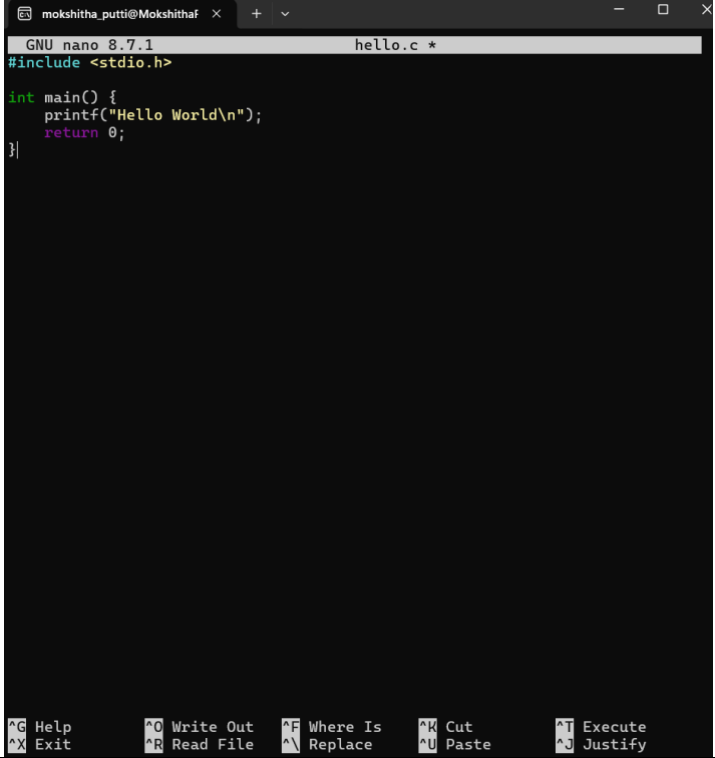
Topic 2: Directory Management

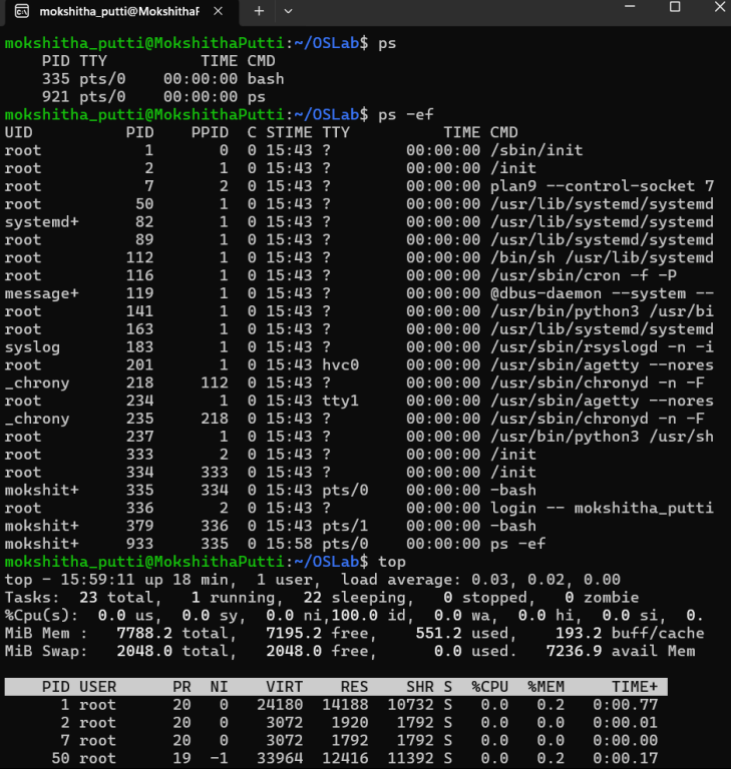
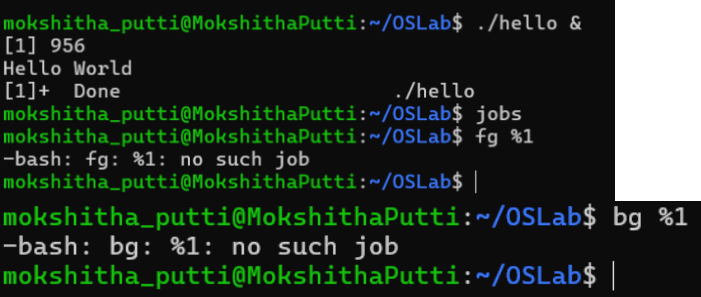
10	ls	List files and directories	ls
11	ls -l	Display detailed file information	ls -l
12	ls -a	Display all files including hidden files	ls -a
13	cd	Change the current directory	cd Documents
14	cd ..	Move to the parent directory	cd ..
15	mkdir	Create a new directory	mkdir OSLab
16	rmdir	Remove an empty directory	rmdir Test

17	find	Search for files and directories	find . -name "*.c"
18	locate	Quickly locate files	locate hello.c
		<pre> mokshitha_putti@MokshithaPutti:~\$ ls ANU hello shell week13.c week16 week19 Week1 hello.c week week14 week16.c week19.c Week1.c s2 week12 week14.c week17 week17.c Week12.c s2example.txt week12.c week15 week17.c week18.c example.txt s2exampleprogram.c.save week13 week15.c week18.c mokshitha_putti@MokshithaPutti:~\$ ls -l total 220 drwxr-xr-x 2 mokshitha_putti mokshitha_putti 4096 Jul 28 04:13 ANU -rwxr-xr-x 1 mokshitha_putti mokshitha_putti 15952 Aug 4 08:45 Week1 -rw-r--r-- 1 mokshitha_putti mokshitha_putti 77 Aug 4 08:45 Week1.c -rw-r--r-- 1 mokshitha_putti mokshitha_putti 158 Jul 31 09:57 Week12.c -rw-r--r-- 1 mokshitha_putti mokshitha_putti 0 Jul 31 04:01 example.txt -rwxr-xr-x 1 mokshitha_putti mokshitha_putti 15952 Aug 4 03:49 hello -rw-r--r-- 1 mokshitha_putti mokshitha_putti 80 Aug 4 03:49 hello.c drwxr-xr-x 2 mokshitha_putti mokshitha_putti 4096 Jul 31 04:02 s2 -rw-r--r-- 1 mokshitha_putti mokshitha_putti 77 Jul 31 09:19 s2example.tx t -rw----- 1 mokshitha_putti mokshitha_putti 77 Jul 31 09:16 s2examplepro gram.c.save -rwxr-xr-x 1 mokshitha_putti mokshitha_putti 16080 Jul 31 15:06 shell -rw-r--r-- 1 mokshitha_putti mokshitha_putti 0 Aug 4 09:03 week -rwxr-xr-x 1 mokshitha_putti mokshitha_putti 16056 Aug 4 08:58 week12 -rw-r--r-- 1 mokshitha_putti mokshitha_putti 158 Aug 4 08:57 week12.c -rwxr-xr-x 1 mokshitha_putti mokshitha_putti 16000 Aug 4 08:59 week13 -rw-r--r-- 1 mokshitha_putti mokshitha_putti 110 Aug 4 08:59 week13.c -rwxr-xr-x 1 mokshitha_putti mokshitha_putti 15992 Aug 4 09:01 week14 -rw-r--r-- 1 mokshitha_putti mokshitha_putti 104 Aug 4 09:01 week14.c -rwxr-xr-x 1 mokshitha_putti mokshitha_putti 15992 Aug 4 09:02 week15 -rw-r--r-- 1 mokshitha_putti mokshitha_putti 230 Aug 4 09:02 week15.c -rwxr-xr-x 1 mokshitha_putti mokshitha_putti 16080 Aug 4 09:03 week16 -rw-r--r-- 1 mokshitha_putti mokshitha_putti 287 Aug 4 09:03 week16.c -rwxr-xr-x 1 mokshitha_putti mokshitha_putti 16032 Aug 4 09:04 week17 -rw-r--r-- 1 mokshitha_putti mokshitha_putti 274 Aug 4 09:04 week17.c -rw-r--r-- 1 mokshitha_putti mokshitha_putti 161 Aug 4 09:08 week18.c -rwxr-xr-x 1 mokshitha_putti mokshitha_putti 16080 Aug 4 09:10 week19 -rw-r--r-- 1 mokshitha_putti mokshitha_putti 264 Aug 4 09:10 week19.c mokshitha_putti@MokshithaPutti:~\$ ls -a . .motd_shown hello.c week13 week17 .. .profile s2 week13.c week17.c .bash_history ANU s2example.txt week14 week18.c mokshitha_putti@MokshithaPutti:~\$ ls -a . .motd_shown example.txt week12 week16 .. .profile hello week12.c week16.c .bash_history ANU hello.c week13 week17 .bash_logout Demo s2 week13.c week17.c .bashrc OSLab s2example.txt week14 week18.c .cache Week1 s2exampleprogram.c.save week14.c week19 .config Week1.c shell week15 week19.c .local Week12.c week week15.c mokshitha_putti@MokshithaPutti:~\$ cd Documents -bash: cd: Documents: No such file or directory mokshitha_putti@MokshithaPutti:~\$ cd OSLab mokshitha_putti@MokshithaPutti:~/OSLab\$ cd .. mokshitha_putti@MokshithaPutti:~\$ mkdir Demo mkdir: Demo: File exists mokshitha_putti@MokshithaPutti:~\$ rmdir Test rmdir: failed to remove 'Test': No such file or directory mokshitha_putti@MokshithaPutti:~\$ find . -name "*.c" ./week19.c ./Week12.c ./week15.c ./week13.c ./week16.c ./week18.c ./week14.c ./week17.c ./Week1.c ./week12.c ./hello.c mokshitha_putti@MokshithaPutti:~\$ locate hello.c mokshitha_putti@MokshithaPutti:~\$ find ~ -name "Week1.c" /home/mokshitha_putti/Week1.c mokshitha_putti@MokshithaPutti:~\$ </pre>	
Topic 3: File Management			
19	touch	Create a new empty file	touch file1.txt

20	cat	Display file contents	cat file1.txt
21	cp	Copy files or directories	cp a.txt b.txt
22	mv	Move or rename files	mv old.txt new.txt
23	rm	Delete a file	rm test.txt
24	head	Display first lines of a file	head hello.c
25	tail	Display last lines of a file	tail hello.c
26	more	View a file page by page	more file.txt
27	less	Scroll through a file	less hello.c
28	wc	Count lines, words and characters	wc hello.c
29	diff	Compare two files	diff a.txt b.txt
30	sort	Sort file contents	sort data.txt
31	uniq	Remove duplicate lines	uniq data.txt
32	grep	Search for text inside a file	grep main hello.c

		<pre> mokshitha_putti@MokshithaPutti:~\$ mkdir OSLab mkdir: OSLab: File exists mokshitha_putti@MokshithaPutti:~\$ cd OSLab mokshitha_putti@MokshithaPutti:~/OSLab\$ pwd /home/mokshitha_putti/OSLab mokshitha_putti@MokshithaPutti:~/OSLab\$ mkdir Practical mokshitha_putti@MokshithaPutti:~/OSLab\$ mkdir Skill mokshitha_putti@MokshithaPutti:~/OSLab\$ ls Practical Skill data.txt file1.txt file2.txt test.txt mokshitha_putti@MokshithaPutti:~/OSLab\$ cd Practical mokshitha_putti@MokshithaPutti:~/OSLab/Practical\$ cd .. mokshitha_putti@MokshithaPutti:~/OSLab\$ mkdir Test mokshitha_putti@MokshithaPutti:~/OSLab\$ find . -name "*.c" mokshitha_putti@MokshithaPutti:~/OSLab\$ touch file1.txt mokshitha_putti@MokshithaPutti:~/OSLab\$ cat file1.txt Hello Linux mokshitha_putti@MokshithaPutti:~/OSLab\$ cp file1.txt file2.txt mokshitha_putti@MokshithaPutti:~/OSLab\$ mv file2.txt newfile.txt mokshitha_putti@MokshithaPutti:~/OSLab\$ head file1.txt Hello Linux mokshitha_putti@MokshithaPutti:~/OSLab\$ tail file1.txt Hello Linux mokshitha_putti@MokshithaPutti:~/OSLab\$ more file1.txt Hello Linux mokshitha_putti@MokshithaPutti:~/OSLab\$ less file1.txt mokshitha_putti@MokshithaPutti:~/OSLab\$ wc file1.txt 1 2 12 file1.txt mokshitha_putti@MokshithaPutti:~/OSLab\$ sort file1.txt Hello Linux mokshitha_putti@MokshithaPutti:~/OSLab\$ uniq file1.txt Hello Linux mokshitha_putti@MokshithaPutti:~/OSLab\$ grep Hello file1.txt Hello Linux mokshitha_putti@MokshithaPutti:~/OSLab\$ rm newfile.txt mokshitha_putti@MokshithaPutti:~/OSLab\$ mokshitha_putti@MokshithaPutti:~/OSLab\$ diff a.txt b.txt diff: a.txt: No such file or directory diff: b.txt: No such file or directory mokshitha_putti@MokshithaPutti:~/OSLab\$ echo "Hello" > a.txt mokshitha_putti@MokshithaPutti:~/OSLab\$ echo "Hi" > b.txt mokshitha_putti@MokshithaPutti:~/OSLab\$ ^C mokshitha_putti@MokshithaPutti:~/OSLab\$ diff a.txt b.txt 1c1 < Hello --- > Hi mokshitha_putti@MokshithaPutti:~/OSLab\$ </pre>	
Topic 4: File Editing			
33	nano	Create or edit a file	nano hello.c
34	vi	Edit a file using Vim editor	vi hello.c
		<pre> mokshitha_putti@MokshithaPutti:~/OSLab\$ rm newfile.txt mokshitha_putti@MokshithaPutti:~/OSLab\$ nano hello.c mokshitha_putti@MokshithaPutti:~/OSLab\$ vi hello.c mokshitha_putti@MokshithaPutti:~/OSLab\$ nano hello.c mokshitha_putti@MokshithaPutti:~/OSLab\$ nano hello.c mokshitha_putti@MokshithaPutti:~/OSLab\$ gcc hello.c -o hello mokshitha_putti@MokshithaPutti:~/OSLab\$./hello Hello World mokshitha_putti@MokshithaPutti:~/OSLab\$ </pre>	

			
Topic 5: C Program Compilation and Execution			
35	gcc	Compile a C program	gcc hello.c -o hello
36	./program	Execute the compiled program	./hello
37	which	Locate the executable of a command	which gcc
38	whereis	Locate binary, source, and manual pages	whereis gcc
39	man	Display manual pages	man gcc
		<pre> mokshitha_putti@MokshithaPutti:~/OSLab\$ rm newfile.txt mokshitha_putti@MokshithaPutti:~/OSLab\$ nano hello.c mokshitha_putti@MokshithaPutti:~/OSLab\$ vi hello.c mokshitha_putti@MokshithaPutti:~/OSLab\$ nano hello.c mokshitha_putti@MokshithaPutti:~/OSLab\$ nano hello.c mokshitha_putti@MokshithaPutti:~/OSLab\$ gcc hello.c -o hello mokshitha_putti@MokshithaPutti:~/OSLab\$./hello Hello World mokshitha_putti@MokshithaPutti:~/OSLab\$ mokshitha_putti@MokshithaPutti:~/OSLab\$ which gcc /usr/bin/gcc mokshitha_putti@MokshithaPutti:~/OSLab\$ whereis gcc gcc: /usr/bin/gcc /usr/lib/gcc /usr/libexec/gcc /usr/share/gcc /usr/share/man/man1/gcc.1.gz mokshitha_putti@MokshithaPutti:~/OSLab\$ man gcc mokshitha_putti@MokshithaPutti:~/OSLab\$ </pre>	
Topic 6: Process Management			
40	ps	Display running processes	ps
41	ps -ef	Display all running processes	ps -ef
42	top	Monitor running processes	top
43	kill	Terminate a process	kill 1234

44	kill -9	Forcefully terminate a process	kill -9 1234
			
Topic 7: Job Control			
45	jobs	Display background jobs	jobs
46	bg	Resume a job in the background	bg %1
47	fg	Bring a job to the foreground	fg %1
48	&	Run a process in the background	./hello &
			
Topic 8: File Permissions and Links			
49	chmod	Change file permissions	chmod +x script.sh
50	chmod 755	Set read, write, and execute permissions	chmod 755 script.sh

51	ln	Create a hard link	ln file1 file2
52	ln -s	Create a symbolic link	ln -s file1 shortcut
		<pre> mokshitha_putti@MokshithaPutti:~/OSLab\$ chmod +x hello mokshitha_putti@MokshithaPutti:~/OSLab\$ chmod 755 hello mokshitha_putti@MokshithaPutti:~/OSLab\$ ln hello hello_hard mokshitha_putti@MokshithaPutti:~/OSLab\$ ln -s hello hello_link mokshitha_putti@MokshithaPutti:~/OSLab\$ </pre>	
Topic 9: Disk and Memory Information			
53	df -h	Display disk space usage	df -h
54	du -sh	Display directory size	du -sh .
55	free -h	Display memory usage (Linux)	free -h
Topic 10: User and Environment Management			
56	id	Display user and group IDs	id
57	env	Display environment variables	env
58	export	Set an environment variable	export PATH=\$P ATH:.
59	alias	Create a command shortcut	alias ll='ls -l'
		<pre> mokshitha_putti@MokshithaPutti:~/OSLab\$ df -h Filesystem Size Used Avail Use% Mounted on none 3.9G 0 3.9G 0% /usr/lib/modules/6.6.87.2-microsoft-st andard-WSL2 none 3.9G 4.0K 3.9G 1% /mnt/wsl drivers 376G 126G 250G 34% /usr/lib/wsl/drivers /dev/sdd 1007G 2.0G 954G 1% / none 3.9G 80K 3.9G 1% /mnt/wslg none 3.9G 0 3.9G 0% /usr/lib/wsl/lib rootfs 3.8G 2.7M 3.8G 1% /init none 3.9G 532K 3.9G 1% /run none 3.9G 0 3.9G 0% /run/lock none 3.9G 0 3.9G 0% /run/shm none 3.9G 76K 3.9G 1% /mnt/wslg/versions.txt none 3.9G 76K 3.9G 1% /mnt/wslg/doc C:\ 376G 126G 250G 34% /mnt/c D:\ 100G 97M 100G 1% /mnt/d none 1.0M 0 1.0M 0% /run/credentials/systemd-journald.serv ice tmpfs 3.9G 0 3.9G 0% /tmp none 1.0M 0 1.0M 0% /run/credentials/systemd-resolved.serv ice none 1.0M 0 1.0M 0% /run/credentials/getty@tty1.service none 1.0M 0 1.0M 0% /run/credentials/console-getty.service tmpfs 779M 0 779M 0% /run/user/1000 mokshitha_putti@MokshithaPutti:~/OSLab\$ du -sh . 40K mokshitha_putti@MokshithaPutti:~/OSLab\$ free -h total used free shared buff/cache avail Mem: 7.6Gi 551Mi 7.0Gi 3.4Mi 200Mi 7 .1Gi Swap: 2.0Gi 0B 2.0Gi mokshitha_putti@MokshithaPutti:~/OSLab\$ </pre>	

		<pre> mokshitha_putti@MokshithaPutti:~/OSLab\$ id uid=1000(mokshitha_putti) gid=1000(mokshitha_putti) groups=1000(mokshitha_putti),4(adm),24(cdrom),27(sudo),30(dip),46(plugdev),100(users) mokshitha_putti@MokshithaPutti:~/OSLab\$ env SHELL=/bin/bash WSL2_GUI_APPS_ENABLED=1 WSL_DISTRO_NAME=Ubuntu NAME=MokshithaPutti PWD=/home/mokshitha_putti/OSLab LOGNAME=mokshitha_putti HOME=/home/mokshitha_putti LANG=C.UTF-8 WSL_INTEROP=/run/WSL/334_interop LS_COLORS=rs=0:di=01;34:ln=01;36:mh=00:pi=40;33:so=01;35:do=01;35:bd=40;33;0 1:cd=40;33;01:or=40;31;01:mi=00:su=37;41:sg=30;43:ca=00:tw=30;42:ow=34;42:st =37;44:ex=01;32:*.tar=01;31:*.tgz=01;31:*.arc=01;31:*.arj=01;31:*.taz=01;31: *.lha=01;31:*.lzh=01;31:*.lzh=01;31:*.lzm=01;31:*.tlz=01;31:*.txz=01;31:*.t zo=01;31:*.t7z=01;31:*.zip=01;31:*.z=01;31:*.dz=01;31:*.gz=01;31:*.lrz=01;31 :*.lz=01;31:*.lzo=01;31:*.xz=01;31:*.zst=01;31:*.tzt=01;31:*.bz2=01;31:*.bz =01;31:*.tbz=01;31:*.tbz2=01;31:*.tz=01;31:*.deb=01;31:*.rpm=01;31:*.jar=01; 31:*.war=01;31:*.ear=01;31:*.sar=01;31:*.rar=01;31:*.alz=01;31:*.ace=01;31:*. zoo=01;31:*.cpio=01;31:*.7z=01;31:*.rz=01;31:*.cab=01;31:*.wim=01;31:*.swm =01;31:*.dwm=01;31:*.esd=01;31:*.avif=01;35:*.jpg=01;35:*.jpeg=01;35:*.mjpg=0 1;35:*.mjpeg=01;35:*.gif=01;35:*.bmp=01;35:*.pbm=01;35:*.pgm=01;35:*.ppm=01; 35:*.tga=01;35:*.xbm=01;35:*.xpm=01;35:*.tif=01;35:*.tiff=01;35:*.png=01;35: *.svg=01;35:*.svgz=01;35:*.mng=01;35:*.pcx=01;35:*.mov=01;35:*.mpg=01;35:*.m peg=01;35:*.m2v=01;35:*.mkv=01;35:*.webm=01;35:*.webp=01;35:*.ogm=01;35:*.mp 4=01;35:*.m4v=01;35:*.mp4v=01;35:*.vob=01;35:*.qt=01;35:*.nuv=01;35:*.wmv=01 ;35:*.asf=01;35:*.rm=01;35:*.rmvb=01;35:*.flc=01;35:*.avi=01;35:*.fli=01;35: *.flv=01;35:*.gl=01;35:*.dl=01;35:*.xcf=01;35:*.xwd=01;35:*.yuv=01;35:*.cgm =01;35:*.m4a=00;36:*.mid=00;36:*.midi=00;36:*.mka=00;36:*.mp3=00;36:*.mpc=00;36: *.ogg=00;36:*.ra=00;36:*.wav=00;36:*.oga=00;36:*.opus=00;36:*.spx=00;36:*.xsp f=00;36:*.xspf=00;90:*.xspf=00;90:*.bak=00;90:*.old=00;90:*.orig=00;90:*.part=00;90 :*.rej=00;90:*.swp=00;90:*.tmp=00;90:*.dpkg-dist=00;90:*.dpkg-old=00;90:*.ucf -dist=00;90:*.ucf-new=00;90:*.ucf-old=00;90:*.rpmnew=00;90:*.rpmorig=00;90:*.r pmsave=00;90: WAYLAND_DISPLAY=wayland-0 LESSCLOSE=/usr/bin/lesspipe %s %s TERM=xterm-256color LESSOPEN= /usr/bin/lesspipe %s mokshitha_putti@MokshithaPutti:~/OSLab\$ export PATH=\$PATH:./ mokshitha_putti@MokshithaPutti:~/OSLab\$ alias ll='ls -l' mokshitha_putti@MokshithaPutti:~/OSLab\$ ll total 52 drwxr-xr-x 2 mokshitha_putti mokshitha_putti 4096 Aug 4 15:46 Practical drwxr-xr-x 2 mokshitha_putti mokshitha_putti 4096 Aug 4 15:46 Skill drwxr-xr-x 2 mokshitha_putti mokshitha_putti 4096 Aug 4 15:47 Test -rw-r--r-- 1 mokshitha_putti mokshitha_putti 0 Aug 4 15:08 data.txt -rw-r--r-- 1 mokshitha_putti mokshitha_putti 12 Aug 4 15:47 file1.txt -rwxr-xr-x 2 mokshitha_putti mokshitha_putti 15952 Aug 4 15:57 hello -rw-r--r-- 1 mokshitha_putti mokshitha_putti 78 Aug 4 15:57 hello.c -rwxr-xr-x 2 mokshitha_putti mokshitha_putti 15952 Aug 4 15:57 hello_hard lrwxrwxrwx 1 mokshitha_putti mokshitha_putti 5 Aug 4 16:01 hello_link -> hello -rw-r--r-- 1 mokshitha_putti mokshitha_putti 0 Aug 4 15:08 test.txt mokshitha_putti@MokshithaPutti:~/OSLab\$ </pre>	
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